

UNIT 3: DATABASE MANAGEMENT SYSTEM (DBMS) & SQL

1. Database

Definition:

A database is an organized collection of related data stored in a computer system.

2. Need of Database

- To store large amount of data
- To reduce data redundancy
- To maintain data security
- Easy retrieval of data

3. DBMS (Database Management System)

Definition:

DBMS is a software used to create, manage and manipulate databases.

Examples: MySQL, Oracle

Relational Data Model

In relational model, data is stored in tables.

Important Terms

Domain

Set of possible values for a column.

Tuple

A row in a table.

Relation

A table in a database.

Key

An attribute used to uniquely identify records.

Types of Keys

Primary Key

- Uniquely identifies each record
- Cannot be NULL

Candidate Key

- All possible keys that can uniquely identify a record

Alternate Key

- Candidate key which is not selected as primary key
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Relational Algebra Operations

Selection (σ)

Selects specific rows from a table.

Projection (π)

Selects specific columns from a table.

Union ($\cup$)

Combines rows of two tables.

Cartesian Product ($\times$)

Combines all rows of first table with all rows of second table.

SQL (Structured Query Language)

SQL is used to create and manipulate databases.

Advantages of SQL

- Easy to use
 - Standard language
 - Fast data retrieval
 - Supports large databases
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Types of SQL Commands

DDL (Data Definition Language)

Used to define database structure.

- CREATE
- DROP
- ALTER

DML (Data Manipulation Language)

Used to manipulate data.

- INSERT
- UPDATE
- DELETE
- SELECT

SQL Data Types

- NUMBER / DECIMAL
- CHARACTER / VARCHAR
- DATE

Important SQL Commands

CREATE TABLE

Used to create a table.

DROP TABLE

Used to delete a table.

ALTER TABLE

Used to modify table structure.

INSERT

Used to insert data into table.

UPDATE

Used to modify existing data.

DELETE

Used to remove records.

SELECT Statement

Used to retrieve data.

Clauses:

- SELECT
- DISTINCT
- FROM
- WHERE
- IN
- BETWEEN
- GROUP BY
- HAVING
- ORDER BY

SQL Aggregate Functions

- SUM()
- AVG()
- COUNT()
- MAX()
- MIN()

Join Operations

Equi-Join

Join tables using common column.

Cartesian Join

All rows of first table combine with second table.

Union

Combine rows of two tables.

Important Exam Questions

- Define DBMS
- Primary Key vs Candidate Key
- DDL vs DML

- Selection vs Projection
 - Write SQL query using WHERE clause
 - Write query using GROUP BY and HAVING
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Exam Tips

- Write SQL keywords in CAPITAL LETTERS.
- Write syntax clearly.
- Underline important terms (Primary Key, DDL, etc.).
- Write differences in table format.